

ADVERSE DRUG REACTIONS AND PHARMACOVIGILANCE

A comprehensive academic report based on the video
"Understanding Adverse Drug Reactions & Pharmacovigilance"
by Dr. Nirmal George (DigiNerve)

REPORT DETAILS	
Topic	Adverse Drug Reactions (ADRs) and Pharmacovigilance
Core source	YouTube lecture by Dr. Nirmal George, DigiNerve
Video publication	30 April 2025
Prepared for	Academic / Pharmacovigilance learning and submission

Purpose: To explain the major concepts presented in the lecture and place them in the practical context of drug-safety monitoring, causality assessment, ADR reporting, and the Pharmacovigilance Programme of India (PvPI).

Prepared as a structured study report

Executive Summary

Adverse drug reactions (ADRs) are unintended and harmful responses to medicines at normally used doses. Pharmacovigilance (PV) is the broader science and set of activities used to detect, assess, understand, prevent, and manage adverse effects and other medicine-related problems. The video by Dr. Nirmal George provides a practical overview of these subjects with emphasis on ADR classification, hypersensitivity, teratogenicity, causality assessment, and ADR reporting in India. The lecture is particularly relevant to pharmacy and medical students because it connects pharmacology knowledge with patient-safety practice.

A central idea in pharmacovigilance is that an adverse event is not automatically an ADR. An adverse event is an untoward medical occurrence after exposure to a medicinal product, while an ADR implies a suspected causal relationship with the medicine. Therefore, drug-safety work requires careful collection of patient, drug, event, timing, alternative-cause, and outcome information before a causal conclusion is made.

The report explains three major causality approaches highlighted by the lecture: the WHO-UMC system, the Naranjo algorithm, and the Bradford Hill considerations. It also describes the practical reporting pathway in India through PvPI, including the suspected ADR reporting form and patient-reporting mechanisms. The final sections summarize the role of pharmacovigilance professionals and the importance of high-quality, timely, complete case information.

Key message: good pharmacovigilance begins with recognition and reporting of suspected reactions, followed by systematic clinical assessment and evidence-based safety evaluation.

1. Introduction

Medicines are intended to produce therapeutic benefits, but no medicine is completely free of risk. A medicine may cause unwanted effects because of its pharmacological action, an immune response, genetic variation, an interaction with another substance, an error in use, or other patient- and treatment-related factors. Once a medicine is widely used, uncommon or delayed safety problems may become visible only after exposure in large populations.

Pharmacovigilance therefore serves as an important component of patient care and rational medicine use. WHO describes pharmacovigilance as the science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other medicine-related problem. The WHO Programme for International Drug Monitoring uses individual case safety reports (ICSRs) from participating countries and contributes to international signal detection through Vigibase. [1,2]

The lecture places ADRs within this larger safety system and emphasizes the movement from recognition -> documentation -> reporting -> causality assessment -> signal evaluation -> risk minimization.

2. Learning Objectives

- Define adverse event, adverse drug reaction, and pharmacovigilance.
- Understand major classifications and mechanisms of ADRs.
- Differentiate hypersensitivity reactions and recognize their clinical relevance.
- Understand drug-related teratogenicity and the historical pregnancy-risk categories discussed in the lecture.
- Explain WHO-UMC, Naranjo, and Bradford Hill approaches to causality assessment.

- Describe the practical steps involved in reporting a suspected ADR in India.
- Understand the role of PvPI, ADR Monitoring Centres (AMCs), healthcare professionals, patients, and pharmaceutical companies.

3. Key Terminology

Term	Meaning / Practical interpretation
Adverse Event (AE)	Any untoward medical occurrence after administration of a pharmaceutical product; a causal relationship with the treatment does not have to be established.
Adverse Drug Reaction (ADR)	A noxious, unintended response to a medicine occurring at normally used doses for prophylaxis, diagnosis, or treatment / modification of physiological function.
Serious ADR	An ADR meeting serious outcome criteria such as death, life-threatening condition, hospitalization or prolongation of hospitalization, significant disability, congenital anomaly, or another medically important situation, depending on the applicable reporting framework.
Unexpected ADR	A reaction whose nature, severity, specificity, or outcome is not consistent with the available product safety information.
ICSR	Individual Case Safety Report: a structured report describing a suspected adverse reaction in an individual patient.
Signal	Information suggesting a possible new causal association or a new aspect of a known association that warrants further investigation.
Pharmacovigilance	The science and activities focused on detecting, assessing, understanding, and preventing adverse effects and other medicine-related problems.

The distinction between AE and ADR is especially important. A temporal association alone does not prove causality. PV professionals must assess the clinical sequence and alternative explanations before deciding how strongly the drug is implicated.

4. Classification of Adverse Drug Reactions

The lecture presents several ways of classifying ADRs. Classification helps clinicians and pharmacovigilance professionals anticipate mechanisms, recognize patterns, and select appropriate investigations.

Category / mechanism	Core idea	Typical examples / relevance
Pharmacological	Predictable effect related to known pharmacological action, often dose-related.	Excessive bleeding with anticoagulants; hypoglycaemia with insulin or sulfonylureas.
Pathological / toxic	Tissue or organ injury resulting from drug exposure or toxic effects.	Hepatotoxicity, nephrotoxicity, marrow toxicity.
Allergic / immunological	Immune-mediated reaction in a sensitized individual; exposure may produce reactions unrelated to the intended pharmacological effect.	Urticaria, angioedema, bronchospasm, anaphylaxis.
Idiosyncratic	Unusual and largely unpredictable response, often related to individual susceptibility.	Rare severe reaction not readily explained by ordinary pharmacology.
Pharmacogenetic	Response influenced by genetic variation in drug metabolism, transport, targets, or immune pathways.	Certain drug toxicities associated with enzyme or HLA variants.
Genotoxic / carcinogenic	Damage to genetic material or increased risk of malignancy associated with exposure.	Relevant to long-term safety assessment of particular therapies.
Teratogenic / developmental	Harm to the embryo or fetus following exposure during pregnancy, especially during susceptible developmental periods.	Congenital malformations or other adverse developmental outcomes.

4.1 Predictable versus unpredictable reactions

A useful conceptual distinction is between reactions that are an extension of known pharmacology and reactions that are unusual or immune/genetically mediated. Predictable reactions are often dose-related and may be mitigated through dose adjustment, monitoring, contraindication checks, or prevention strategies. Unpredictable reactions can be harder to anticipate and may require awareness of previous allergy, genetics, immune status, or rare susceptibility factors.

4.2 Why classification matters in pharmacovigilance

- It guides the immediate clinical response and investigation of a suspected reaction.
- It helps determine what patient history and laboratory information should be collected.
- It supports consistent medical coding and case assessment.
- It helps safety teams identify patterns and prioritize possible signals.

5. Hypersensitivity Reactions

The lecture highlights the four classic types of hypersensitivity. These are useful for understanding allergic and immune-mediated ADRs. In practice, a patient can present with overlapping mechanisms, and clinical management should be based on the observed syndrome rather than on a classroom label alone.

Type	Main mechanism	Typical examples / timing concept
Type I - immediate	IgE-mediated mast-cell activation with release of mediators.	Urticaria, allergic rhinitis, bronchospasm, anaphylaxis; often rapid after exposure.
Type II - antibody-mediated	IgG/IgM directed against cell-surface or tissue antigens, with complement or cellular effects.	Drug-induced hemolysis, thrombocytopenia and other cytopenias in selected settings.

Type	Main mechanism	Typical examples / timing concept
Type III - immune complex	Circulating antigen-antibody complexes deposit in tissues and activate inflammatory pathways.	Serum sickness-like presentations and some vasculitic reactions.
Type IV - delayed T-cell-mediated	T-cell-driven inflammatory response without a primary antibody mechanism.	Contact dermatitis and several delayed drug eruptions.

Clinical importance: Severe immediate reactions such as anaphylaxis are medical emergencies. From a PV perspective, accurate documentation should include suspected medicine(s), dose, route, timing, onset, clinical manifestations, treatment, outcome, and relevant previous exposure or allergy history.

6. Teratogenicity and Medicines in Pregnancy

Teratogenicity refers to drug-associated harm to the developing embryo or fetus. Risk depends on factors such as the medicine, dose, timing of exposure, maternal and fetal characteristics, and the developmental process affected. The classic teaching model discussed in the lecture uses pregnancy categories A, B, C, D, and X.

Important current-context note: the A/B/C/D/X letter system was a historical United States labeling approach. The U.S. Food and Drug Administration replaced those letter categories with the Pregnancy and Lactation Labeling Rule (PLLR), which uses narrative risk information. Therefore, the historical categories are useful for understanding the lecture and older academic material, but they should not be treated as the current U.S. labeling framework.

Historical category	Traditional teaching interpretation
A	Controlled human data did not demonstrate a risk under the conditions described.
B	Animal data did not demonstrate risk or showed findings without confirmation in human data; interpretations varied by the specific drug and evidence.
C	Risk could not be ruled out; animal studies may have shown adverse effects and adequate human data were limited.
D	Positive evidence of human fetal risk existed, but potential benefits could justify use in certain clinical circumstances.
X	Evidence indicated fetal risk and potential risks were considered to outweigh possible benefits in pregnancy.

In pharmacovigilance, pregnancy exposures are important because some developmental effects may only become apparent after a substantial time interval. Cases should therefore be followed carefully for pregnancy outcome, congenital anomalies, and relevant maternal/fetal history.

7. Causality Assessment of ADRs

A central pharmacovigilance question is: How likely is it that the suspected medicine caused the event? The answer is rarely based on a single fact. Assessors consider the temporal relationship, dechallenge and rechallenge information when available, alternative causes, known pharmacology, objective evidence, dose-response information, prior evidence, and clinical plausibility.

7.1 WHO-UMC Causality Assessment

The WHO-UMC system uses clinical-pharmacological reasoning and assigns categories such as Certain, Probable/Likely, Possible, Unlikely, Conditional/Unclassified, and Unassessable/Unclassifiable. For example, a “Probable/Likely” case generally has a reasonable time relationship, is unlikely to be explained by disease or other medicines, and shows a clinically reasonable response to withdrawal; rechallenge is not required. [3]

Category	Simplified interpretation
Certain	Clear temporal relationship; alternative explanations unlikely; dechallenge is plausible and the event has a definitive pharmacological/phenomenological basis; rechallenge may strengthen the assessment when appropriate.
Probable / Likely	Reasonable time relationship; other explanations are unlikely; response to withdrawal supports the relationship; rechallenge not required.
Possible	Reasonable time relationship, but disease or other drugs could also explain the event; withdrawal information may be incomplete or unclear.
Unlikely	Timing makes the relationship improbable and another cause is more plausible.
Conditional / Unclassified	A suspected reaction is present, but more information or ongoing assessment is required.

Category	Simplified interpretation
Unassessable / Unclassifiable	Information is insufficient or contradictory and cannot reasonably be supplemented or verified.

7.2 Naranjo Algorithm

The Naranjo Adverse Drug Reaction Probability Scale is a structured 10-question approach. Questions address previous reports, temporal sequence, improvement on withdrawal, recurrence on re-exposure, alternative causes, placebo response, toxic concentrations, dose-response, previous similar reactions, and objective evidence. Scores are interpreted as: 9 or more = definite; 5-8 = probable; 1-4 = possible; 0 or less = doubtful. [4,5]

Key question area	Typical scoring direction
Previous conclusive reports?	Yes +1
Event appeared after drug administration?	Yes +2; No -1
Improved after discontinuation / antagonist?	Yes +1
Reappeared after readministration?	Yes +2; No -1
Alternative causes present?	Yes -1; No +2
Reappeared with placebo?	Yes -1; No +1
Toxic concentration documented?	Yes +1
Dose-response present?	Yes +1
Similar reaction previously with same/similar drug?	Yes +1
Objective evidence?	Yes +1

7.3 Bradford Hill Considerations

The Bradford Hill considerations were originally developed for assessing causal relationships in epidemiology. UMC describes nine commonly used considerations: strength of association, consistency, specificity, temporality, dose-response, biological plausibility, coherence, experimental evidence, and analogy. Reversibility is sometimes considered as an additional practical factor. These considerations are particularly useful when thinking about population-level evidence and signals rather than simply scoring one patient case. [6]

8. ADR Reporting and the Pharmacovigilance Process

Reporting is the starting point of spontaneous pharmacovigilance. A suspected reaction does not need to be proven before it is reported. The goal is to collect enough reliable information for subsequent medical review, follow-up, causality assessment, and signal detection.

8.1 Minimum information and good-quality case reporting

- A patient who can be identified without exposing unnecessary personal information.
- A reporter who can be identified and contacted for clarification or follow-up.
- A suspected medicinal product.
- Description of one or more suspected adverse events / reactions with relevant dates and clinical details.
- Additional information such as dose, route, indication, concomitant medicines, medical history, investigations, treatment of the reaction, dechallenge/rechallenge, and outcome whenever available.

The quality of a case report often determines the quality of the subsequent safety assessment. Incomplete dates, missing dose information, absent concomitant medication history, or vague event descriptions can make causality assessment and signal detection more difficult.

8.2 Reporting in India: Pharmacovigilance Programme of India (PvPI)

The Pharmacovigilance Programme of India (PvPI), coordinated through the Indian Pharmacopoeia Commission, is the national pharmacovigilance framework referenced by the lecture. PvPI encourages healthcare professionals, patients, and other relevant reporters to submit suspected adverse events. Reports can be submitted using the Suspected ADR Reporting Form, through ADR Monitoring Centres, and through available patient reporting mechanisms. [7-9]

Actor	Role in the safety system
Patient / consumer	Recognizes and communicates suspected medicine-related problems; provides medication and event details.
Doctor / clinician	Recognizes, treats, documents, and reports suspected ADRs; provides clinical interpretation.
Pharmacist / nurse / other HCP	Identifies suspected reactions, supports medication history and reporting, and contributes to patient safety.
ADR Monitoring Centre (AMC)	Receives, reviews, follows up, and forwards/handles ADR reports within the PV system.
NCC-PvPI / IPC	Coordinates the national programme, evaluates aggregate information and supports signal and safety activities.
Pharmaceutical company	Maintains product safety surveillance and submits relevant safety information/ICSRs according to applicable regulatory requirements.

8.3 Practical ADR Reporting Workflow

1. Recognize: identify a suspected drug-event relationship. 2. Stabilize: ensure the patient receives appropriate clinical care. 3. Document: record drug, dose, indication, timing, event description and outcome. 4. Report: submit the suspected ADR through the appropriate PvPI mechanism / AMC. 5. Follow up: obtain missing information and updated outcomes where possible. 6. Assess: perform clinical review and causality assessment. 7. Aggregate and detect signals: look across cases for patterns that could indicate a new or changed safety risk. 8. Manage risk: communicate or implement appropriate risk-minimization measures when supported by the evidence.

9. Pharmacovigilance Signal Detection

Signal detection is the process of identifying evidence that a medicine may be associated with an adverse event in a way that is new, stronger, or otherwise different from what was previously understood. Signals are not automatically proof of causation; they are prompts for further investigation. PvPI guidance notes that case series can be valuable for hypothesis generation, particularly for unusual serious events, while broader analyses and international data can strengthen or weaken a suspected association. [10]

Examples of information that can contribute to signal detection include spontaneous reports, literature, clinical and epidemiological studies, regulatory databases, product quality complaints, and other post-marketing safety data. Modern pharmacovigilance therefore combines individual clinical case review with aggregate and population-level evidence.

10. Practical Example of ADR Assessment

Hypothetical case: A patient starts a new medicine. Two days later, the patient develops a widespread itchy rash. The medicine is stopped, the rash improves, and no new medicine is introduced during the same period. The patient had no similar rash before treatment, and there is a plausible timing relationship.

Step 1 - Verify the event: document the rash, onset date, severity, treatment, and outcome.

Step 2 - Verify exposure: confirm the suspected medicine, dose, route, indication, start date, and adherence.

Step 3 - Examine alternatives: assess other medicines, infections, food exposures, and underlying conditions that could explain the rash.

Step 4 - Assess dechallenge: the improvement after withdrawal supports a relationship, although it does not by itself prove causation.

Step 5 - Apply a structured method: use WHO-UMC clinical reasoning or the Naranjo questions, depending on the setting and purpose.

Step 6 - Report: submit the suspected ADR even when certainty is not established, because spontaneous reports are intended to generate safety knowledge.

Step 7 - Follow up: obtain additional clinical information, relevant laboratory findings, treatment, and final outcome if available.

Lesson: a good pharmacovigilance case is not merely a statement that “the drug caused the rash.” It is a documented chain of evidence that allows another reviewer to reconstruct what happened and evaluate alternative explanations.

11. Importance of Pharmacovigilance in Public Health

- Protects patients by identifying important medicine-related risks that may not be fully characterized before widespread use.
- Improves the risk-benefit assessment of medicines throughout their life cycle.
- Supports safer prescribing, dispensing, monitoring, and patient counselling.
- Helps identify rare or delayed reactions through large-scale exposure data.
- Allows regulators and manufacturers to introduce warnings, contraindications, monitoring recommendations, or other risk-minimization measures when justified.
- Contributes to rational medicine use and confidence in healthcare systems.

WHO emphasizes that pharmacovigilance is not limited to detecting side effects after marketing. Its wider scope includes identifying and evaluating medicine-related problems and supporting patient safety and public health. [1,2]

12. Role of a Pharmacovigilance Professional

For a pharmacy graduate entering pharmacovigilance, the subject combines pharmaceutical knowledge, clinical understanding, scientific writing, structured data review, and attention to detail. Typical responsibilities across industry and PV service organizations can include case intake, validity checks, data entry, medical coding, narrative preparation, seriousness and expectedness assessment, follow-up, quality review, literature-related work, aggregate reporting, and

signal-related activities, depending on the role.

The most transferable habits are accuracy, consistency, evidence-based reasoning, documentation discipline, and respect for patient confidentiality. A PV professional should avoid both under-reporting and over-interpreting: suspected cases should be captured appropriately, while conclusions should remain proportional to the evidence.

13. Key Takeaways from the Lecture

Concept	What to remember
ADR	An unintended and harmful response occurring at normally used doses; the concept includes a suspected causal relationship.
PV	A broad safety discipline covering detection, assessment, understanding and prevention of medicine-related harm.
ADR classification	Mechanisms can be pharmacological, pathological/toxic, allergic, idiosyncratic, pharmacogenetic, genotoxic, or teratogenic.
Hypersensitivity	Types I-IV describe different immune mechanisms and timing patterns.
Teratogenicity	Pregnancy exposure requires special risk assessment; historical A-X categories should be understood in context.
WHO-UMC	Clinical-pharmacological causality categories such as Certain, Probable/Likely, Possible and Unlikely.
Naranjo	A 10-question scored probability scale: ≥ 9 definite, 5-8 probable, 1-4 possible, ≤ 0 doubtful.
Bradford Hill	Causal considerations useful especially when interpreting epidemiologic and signal-level evidence.
PvPI	India's national pharmacovigilance framework includes ADR reporting pathways, AMCs, national coordination and patient reporting.
Reporting principle	A suspected ADR should be reported with accurate, complete, clinically useful information; proof of causality is not required before reporting.

14. Conclusion

The video “Understanding Adverse Drug Reactions & Pharmacovigilance” provides a concise bridge between pharmacology and real-world patient safety. Its emphasis on ADR recognition, classification, hypersensitivity, teratogenicity, causality assessment, and reporting reflects the fundamental workflow of pharmacovigilance. The key professional lesson is that medicine safety depends on a continuous cycle of observation, reporting, assessment, evidence generation, and risk management.

For students and early-career pharmacovigilance professionals, mastering these fundamentals is valuable because individual case reports form the raw material for larger safety assessments. A well-documented suspected ADR can contribute to the recognition of patterns that may otherwise remain hidden. Conversely, poor-quality or incomplete information can limit the usefulness of a safety report. Accurate documentation, timely reporting, careful causality reasoning, and appropriate follow-up are therefore essential elements of good pharmacovigilance practice.

Overall, ADR monitoring should be viewed not simply as a form-filling activity but as a patient-safety function. The goal is not only to identify that a reaction occurred, but to understand whether and how a medicine contributed to it, whether similar cases are occurring, what the implications are for the benefit-risk balance, and what action may reduce future harm.

15. References and Source Notes

Primary video source

[V1] DigiNerve / Dr. Nirmal George. Understanding Adverse Drug Reactions & Pharmacovigilance. YouTube, published 30 April 2025. <https://youtu.be/jaPje7Pblo0>

Authoritative and supporting sources

[1] World Health Organization (WHO). Programme for International Drug Monitoring. Pharmacovigilance definition, aims and VigiBase / ICSR framework. <https://www.who.int/teams/regulation-prequalification/regulation-and-safety/pharmacovigilance/networks/pidm>

[2] Indian Pharmacopoeia Commission (IPC). Pharmacovigilance Programme of India (PvPI) - Frequently Asked Questions / scope and definitions. <https://ipc.gov.in/PvPI/faq.html>

[3] Uppsala Monitoring Centre (WHO-UMC). WHO-UMC Causality Categories. https://who-umc.org/media/u4gjgxvv/who-umc-causality-assessment_new-logo.pdf

[4] NCBI Bookshelf / LiverTox. Naranjo Adverse Drug Reaction Probability Scale. <https://www.ncbi.nlm.nih.gov/books/NBK548069/>

[5] NCBI. Naranjo Adverse Drug Reaction Probability Scale worksheet. <https://www.ncbi.nlm.nih.gov/books/NBK548069/bin/Naranjoassessment.pdf>

[6] Uppsala Monitoring Centre. Bradford Hill Criteria. <https://edit.who-umc.org/signal-management/bradford-hill-criteria/>

[7] Indian Pharmacopoeia Commission. PvPI ADR Reporting page and reporting forms. <https://www.ipc.gov.in/PvPI/adr.html>

[8] Indian Pharmacopoeia Commission. ADR Reporting Forms. <https://www.ipc.gov.in/mandates/pvpi/adr-reporting/8-category-en/865-adr-reporting-forms.html>

[9] CDSCO / IPC. Suspected Adverse Drug Reaction Reporting Form, Version 1.4. https://cdsco.gov.in/opencms/export/sites/CDSCO_WEB/Pdf-documents/Consumer_Section_PDFs/ADRRF_2.pdf

[10] Indian Pharmacopoeia Commission. PvPI Toolkit / signal detection and spontaneous reporting guidance. <https://ipc.gov.in/PvPI/toolkit.html>

Academic note: This report is a structured study document based on the identified contents of the linked lecture and supplemented with current authoritative pharmacovigilance sources. It is intended for educational use and should not replace official regulatory guidance, product information, or clinical judgment.

End of Report